EECS 2311 Group 17 - Student Budget and Expense Tracker

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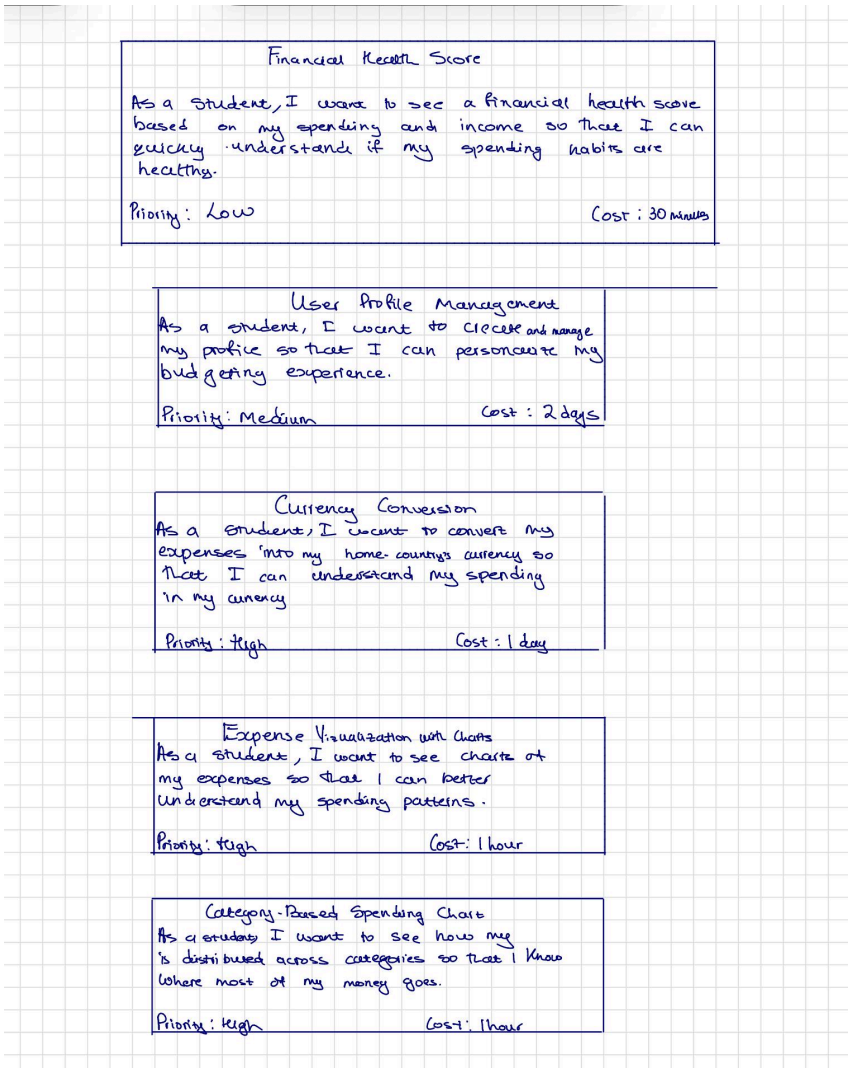
ITER 2 Update

Revised User Stories (Iteration 2)

During this iteration, several user stories were revised and expanded based on the progress of the project and feedback from the instructor. The original stories mainly focused on basic expense tracking and budgeting by academic term. In this iteration, I extended those stories to include additional insights, visualization, personalization, and York-specific financial planning features.

These revisions allow the system to better support how students actually plan their finances during a school term.

Revised Added User Stories:



Financial Health Score

As a student, I want to see a financial health score based on my spending and income so that I can quickly understand if my spending habits are healthy.

Priority: Low

Cost: 30 minutes

User Profile Management

As a student, I want to create and manage my profile so that I can personalize my budgeting experience.

Priority: Medium

Cost: 2 days

Currency Conversion

As a student, I want to convert my expenses into my home-country's currency so that I can understand my spending in my currency

Priority: High

Cost: 1 day

Expense Visualization with Charts

As a student, I want to see charts of my expenses so that I can better understand my spending patterns.

Priority: High

Cost: 1 hour

Category-Based Spending Chart

As a student, I want to see how my is distributed across categories so that I know where most of my money goes.

Priority: High

Cost: 1 hour

York Tuition Cost Estimation

As a York student, I want to estimate my tuition based on my program so that I can plan my expenses.

Priority: High Cost: 1 day

Campus Residence Cost Estimation

As a student living on campus, I want to see the estimated residence cost so that I can include housing in my budget.

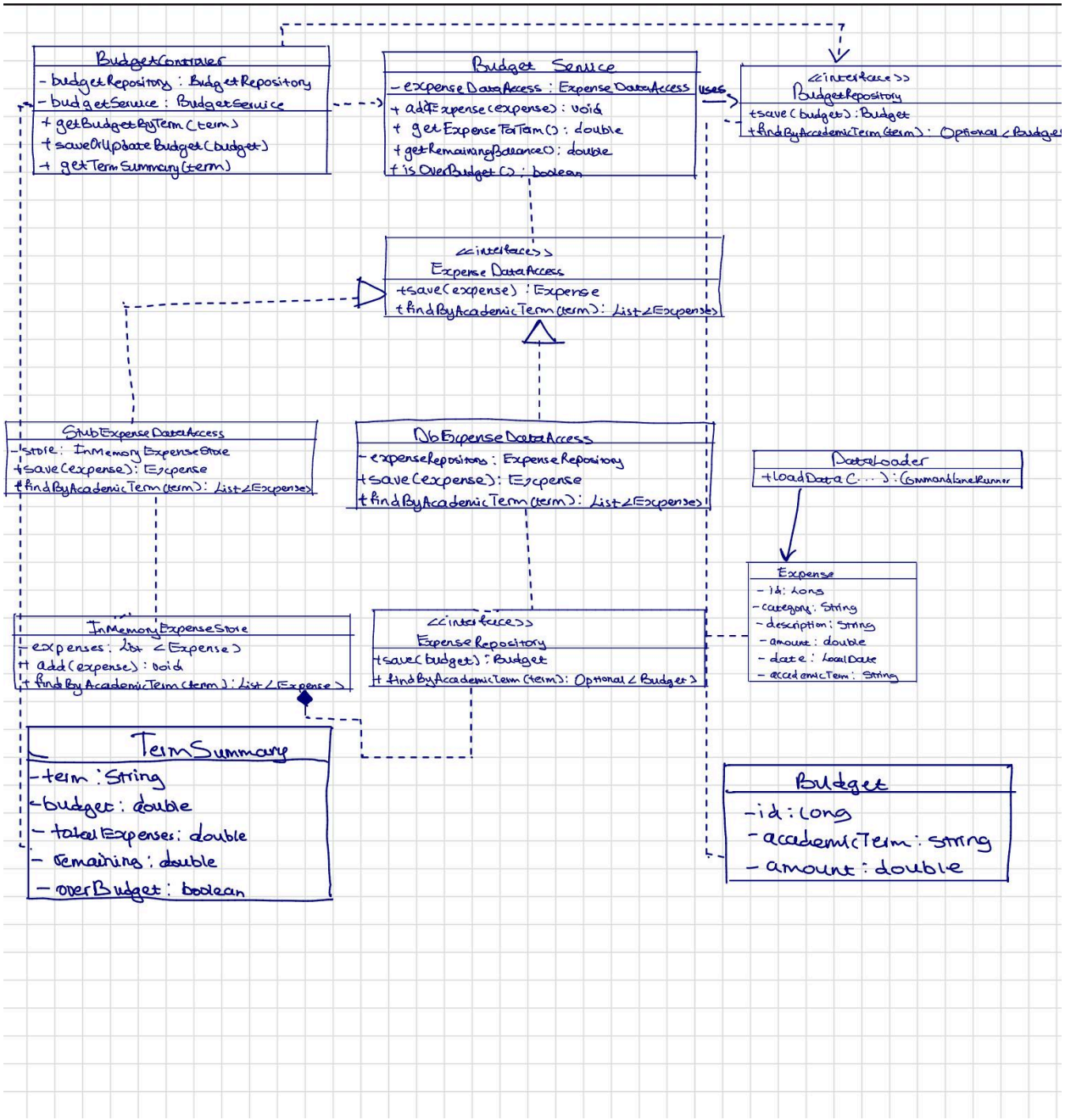
Priority: High Cost: 1 day

Education Cost Breakdown

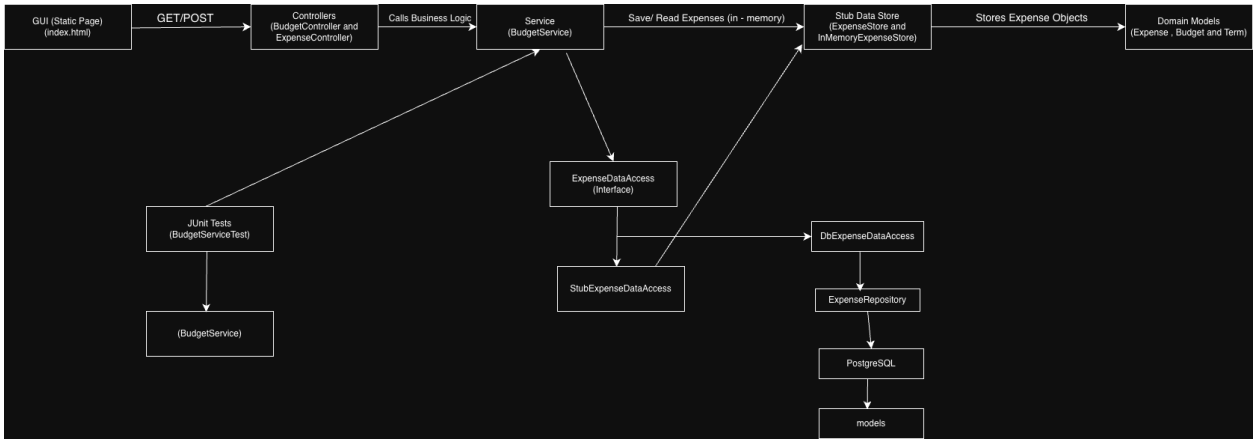
As a student, I want to see a breakdown of education costs so that I can understand where my money is going.

Priority: Medium Cost: 1 day

UML Diagram:


BudgetController
- budgetRepository : BudgetRepository
- budgetService : BudgetService
+ getBudgetByTerm (term)
+ saveOrUpdateBudget (budget)
+ getTermSummary (term)
BudgetService
- expenseDataAccess : ExpenseDataAccess
+ addExpense (expense) : void
+ getExpenseForTerm () : double
+ getRemainingBalance() : double
+ isOverBudget () : boolean
uses
<<interface>>
BudgetRepository
+save (budget) : Budget
+findByAcademicTerm (term) : Optional
<<interface>>
ExpenseDataAccess
+save(expense) : Expense
+findByAcademicTerm (term) : List<Expense>
StubExpenseDataAccess
- store : InMemoryExpenseStore
+save(expense) : Expense
+findByAcademicTerm (term) : List<Expense>
DbExpenseDataAccess
- expenseRepository : ExpenseRepository
+save(expense) : Expense
+findByAcademicTerm (term) : List<Expense>
DataLoader
+loadData (...) : CommandLineRunner
Expense
- id : Long
- category : String
- description : String
- amount : double
- date : LocalDate
- academicTerm : String
InMemoryExpenseStore
- expenses : List<Expense>
+ add (expense) : void
+ findByAcademicTerm (term) : List<Expense>
<<interface>>
ExpenseRepository
+save(budget) : Budget
+findByAcademicTerm (term) : Optional<Budget>
TermSummary
- term : String
- budget : double
- totalExpenses : double
- remaining : double
- overBudget : boolean
Budget
- id : Long
- academicTerm : String
- amount : double

Updated Archictecture Diagram



Added User Stories:

1.) Financial Health Score

As a student, I want to see a financial health score based on my spending and income so that I can quickly understand whether my spending habits are healthy.

Priority: Low
Planned Cost: 30 minutes
Actual Cost: ~45 minutes

This feature calculates a simple score based on how much of the student's budget has been used and whether their spending is approaching the term budget limit.

2.) User Profile Management

As a student, I want to create and manage my profile so that I can personalize my budgeting experience.

Priority: Medium
Planned Cost: 2 days
Actual Cost: ~2 days

The profile allows the system to store information such as preferred currency, student information, and other user settings. This supports other features such as currency conversion.

3.) Currency Conversion

As a student, I want to convert my expenses into my home country's currency so that I can understand my spending in my own currency.

Priority: High
Planned Cost: 1 day
Actual Cost: ~1 day

This feature is especially useful for international students who may want to view their expenses in their home currency instead of Canadian dollars.

4.) Expense Visualization with Charts

As a student, I want to see charts of my expenses so that I can better understand my spending patterns.

Priority: High
Planned Cost: 1 hour
Actual Cost: ~2 hours

Charts allow students to quickly see how their spending is distributed instead of manually reading lists of expenses.

5.) Category-Based Spending Chart

As a student, I want to see how my spending is distributed across categories so that I know where most of my money goes.

Priority: High
Planned Cost: 1 hour
Actual Cost: ~1.5 hours

This chart groups expenses by category and shows a visual breakdown of spending.

6.) York Tuition Cost Estimation

As a York student, I want to estimate my tuition based on my program so that I can plan my academic expenses.

Priority: High
Planned Cost: 1 day
Actual Cost: ~1.5 days

The estimator uses York University tuition data and calculates estimated costs based on faculty, program, and student type (domestic or international).

This feature was added after feedback from the professor during the first presentation to provide a unique feature targeted specifically at York students.

7.) Campus Residence Cost Estimation

As a student living on campus, I want to see the estimated residence cost so that I can include housing in my budget.

Priority: High
Planned Cost: 1 day
Actual Cost: ~2 days

The estimator includes residence building, room type, dining plan, and additional residence fees. It also accounts for residences where dining plans are required versus optional.

8.) Education Cost Breakdown

As a student, I want to see a breakdown of education costs so that I can understand where my money is going.

Priority: Medium
Planned Cost: 1 day
Actual Cost: ~1 day

The breakdown includes tuition, residence costs, meal plans, books and supplies, and additional fees. This helps students understand the total cost of their education rather than only tracking individual expenses.

Changes from the Previous Plan

Compared to the original planning document, the system evolved from a basic expense tracker into a more complete student financial planning tool.

The biggest change was the addition of York-specific cost estimation features, which were not part of the original design but were added after instructor feedback to make the project more unique and directly useful for York students.

The York Cost Estimator feature is still in progress and will continue to be expanded in future iterations. Right now, it supports a few faculties and programs, but the goal is to eventually include all York programs so students can get a more accurate estimate of their tuition. I also plan to add a mode where the estimated tuition and housing costs can automatically feed into the budgeting part of the app, so students can plan their finances more realistically based on their actual school expenses. Another improvement would be allowing students to compare different residence options, meal plans, and credit loads to see how their total costs change. This feature is intended to make the budgeting tool more useful, specifically for York students since it connects real university costs with the budgeting system.

Other revisions mainly focused on improving usability through charts, financial insights, personalization, and currency support.

Student Budget and Expenses Tracker - Vision Statement

The Student Budget and Expenses Tracker is a simple budgeting app made for university students that helps them manage and understand their finances across academic terms rather than traditional monthly budgets.

Most budgeting applications are built around monthly financial cycles. However, university students do not operate on monthly financial patterns. Students typically receive the majority of their funding at the beginning of an academic term (loans, scholarships, family support), and

then spend gradually throughout the semester, with spending spikes during midterms, finals, and major academic deadlines.

This system is designed specifically for undergraduate and graduate students who want a clearer and more realistic way to understand their finances. Instead of tracking spending month-by-month, this application organizes financial activity by academic term (e.g., Fall 2026, Winter 2027).

Students can record expenses and associate them with a specific academic term and category, such as tuition, rent, food, textbooks, subscriptions, and other school-related costs. The system separates high-impact academic expenses from everyday spending so students can clearly see how much of their money is going directly toward school.

The tracker also allows students to plan ahead by listing expected academic expenses before the term begins. This helps students prepare for large costs such as tuition payments, textbooks, lab fees, or course-related materials.

Additionally, the system generates simple, easy-to-understand summaries that can be shared with financial advisors, sponsors, or family members. This makes it easier for students who rely on external support to communicate how their money is being used during the term.

The project will be considered successful if students can easily record their expenses by academic term, clearly see how much they've spent, and understand how much money they have left for the semester. The system should show when a student is going over budget, separate major school costs from everyday spending, and run smoothly without errors. Overall, the goal is for students to actually feel more in control of their money during the term.

Big Stories

Term - Based Expense Tracking

As a student I want to record and view my expenses by academic term.

Priority: High Cost: 8 days

Term Budgeting and Spending Insights

As a student I want to set a budget for an academic term and see how much I have spent.

Priority: High Cost: 7 days

Student Focused Planning and Goals

As a student I want to mark important academic expenses and set saving goals during a term.

Priority: Medium Cost: 6 days

Term Review and Advisor Reporting

As a student I want to compare my spending across different academic terms and generate summaries.

Priority: Medium Cost: 6 days

User Stories.

Add expense to selected academic term

As a student, I want to add a new expense to selected academic term, so that I can accurately track my spending during that semester.

Acceptance Criteria
- The user can enter:
  ① amount ② description
  ③ Category ④ Date ⑤ Academic term
- The expense is saved in the system
- The expense appears when viewing the term
- Amount > 0.

Priority: High Cost: 3 days

Assign Category To An Expense

As a student, I want to assign a predefined category to each expense.

Acceptance Criteria:
- There is a predefined category
- A category must be selecting when adding an expense.
- Expenses can be filtered by category

Priority: High Cost: 1.5 days

Delete Expense

As a student I want to delete an expense so that I can remove accidental entries.

Acceptance Criteria
- The user can select an expense to delete.
- The expense is removed
- The term total updates after deletion

Priority: Medium Cost: 1 day

Edit Existing Expense

As a student I want to edit an expense I have added.

Acceptance:

- The user can modify fields and save changes
- The updated values replace the old values
- Invalid edits are rejected.

Priority: Medium Cost: 1.5 days

View Expenses For a Selected Term

As a student, I want to view all recorded expenses within a term.

Acceptance Criteria

- Expenses are displayed
- Only expenses for the selected term are displayed

Priority: High Cost: 2 days.

Big User Stories

Big Story 1 – Term-based expense tracking

As a student, I want to record and view my expenses by academic term, so that I can clearly understand how I spend my money during each semester.

Priority: High
Cost: 8 days

Big Story 2 – Term budgeting and spending insights

As a student, I want to see summaries of my spending and whether I am staying within my term budget, so that I can manage my money and avoid running out before the semester ends.

Priority: High
Cost: 7 days

Big Story 3 – Student-focused planning and goals

As a student, I want to plan and track important school-related expenses and savings goals during a term, so that I can prepare for large academic costs like tuition and textbooks.

Priority: Medium
Cost: 6 days

Big Story 4 – Multi-term review and reporting

As a student, I want to review and compare my spending across different academic terms and generate simple reports, so that I can reflect on my habits and share summaries with an advisor or family member.

Priority: Medium
Cost: 6 days

PROJECT SCOPE SECTION

Unique customized features/ selling points:

Term cash-flow view (not just totals)
Students can see how their money is spread across the term (early term vs midterm season vs exam period), so they can clearly tell when their spending spikes during busy school weeks.

Big school expense planning
The app lets students mark certain expenses as *planned academic costs* (like tuition, books, lab fees), so these don't feel like random expenses and can be prepared for ahead of time.

International student support mode
Students can tag expenses as "school country" vs "home country" spending (for example travel, remittances, or exchange-related costs), which helps international students better understand their real term budget.

Term survival indicator
Instead of only saying "over budget", the system can show a simple message like:
"At this rate, your money may not last until the end of the term."

This directly matches how students think (“Will I survive this semester?”).

Advisor-friendly report view
The export summary is formatted specifically so a student can easily show it to a financial advisor, sponsor, or school support office without needing to explain the data.

Course-related spending insight
Students can optionally link expenses to a course (for example, textbooks, lab kits, or online tools), so they can see which courses are costing them the most outside tuition.

High-level feature categories (Big User Stories)

1. Term-based expense tracking (ITR1)

Big story

As a student, I want to record and view my expenses by academic term, so that I can clearly understand how I spend my money during each semester.

Smaller user stories

- Add a new expense to a selected academic term (Iyinoluwa Blossom Olu-Alabi)
    - The system must allow a user to select an active academic term (e.g., "Winter 2026") and input a new expense entry. The entry must include a numeric amount, a description (e.g., "Math Lab Kit"), and a specific date within the term's range.
    - Unanswered questions:
        - Should the system allow users to enter expenses for dates that fall outside the defined term start/end dates?
        - Will the system support multiple currencies if the student is an international student?

- Assign a category to an expense (Ayesh Ahmed)
    - When creating or editing an expense, the user can select a category from a predefined list (Tuition, Rent, Food, Textbooks, Subscriptions, etc.) to enable future reporting.
    - Unanswered questions:
        - Can a user add "Custom Categories" if the defaults don't fit their needs?
        - Can one expense have multiple categories?

- Edit or remove an existing expense (Arvin Gholipoor)

    - Users need a way to correct data entry errors or delete accidental duplicates. Selecting an expense from a list should open a form with its current data pre-populated for modification.
    - Unanswered questions:
        - Should users be prompted to confirm deletions to avoid accidental data loss?
        - Should removed expenses be held somewhere temporarily for easy recovery?

- View all expenses for a selected term (Dexter Sargent)
    - The user should be able to see a chronological list of all recorded expenses within a chosen academic term. This view should display the total sum of these expenses at the bottom.
    - Unanswered questions:
        - Should the list be sorted by date (newest first) or by the highest dollar amount by default?
        - Should the user be able to change the way that options are sorted and filtered?
        - Should there be a search bar?

- View expenses grouped by category for a term (Dexter Sargent)
    - This view should aggregate all spending by category (e.g., seeing a single total for "Food") rather than a list of individual transactions.
    - Unanswered questions:
        - Should this be a text-based list or include a basic visual chart (like a progress bar) for each category?
        - How do we handle categories that have $0 in spending? Should they be hidden or shown as empty?

2. Term budgeting and spending insights (ITR2)

Big story

As a student, I want to see summaries and insights for my spending in a term, so that I can manage my money and avoid running out before the semester ends.

Smaller user stories inside this category

- View total spending for a selected academic term

- Enter a planned term budget or income

- See remaining balance for the term
- Receive an over-budget warning when spending exceeds the term budget
- See a simple spending breakdown for the term (by category)

3. Student-focused planning and goals (ITR3)

Big story

As a student, I want to plan and track important school-related costs and goals during a term, so that I can prepare for large academic expenses.

Smaller user stories inside this category

- Mark expenses as major school costs (tuition, textbooks, lab fees, online tools, etc.)
- Set a savings goal for an academic term
- Track progress toward the term savings goal
- View upcoming planned academic expenses within a term

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4. Multi-term review and reporting (ITR3)

Big story

As a student, I want to review and compare my spending across different academic terms and export summaries, so that I can reflect on my habits and share them with an advisor or family member.

Smaller user stories inside this category

- Compare total spending between two academic terms
- View category-level differences between terms
- Export a term summary report
- Generate an advisor-friendly report view